

Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

23CSE498 **Project Phase II** **2026**

B.Tech CSE — Initial Project Idea Submission

Project Team

Roll Number	Name
AM.SC.U4CSE23304	Advyth Vaman Akalankam
AM.SC.U4CSE23331	Madana Lakshmi Tanmayi
AM.SC.U4CSE23344	Rayudu Vamsi Krishna
AM.SC.U4CSE23351	Sangamreddi Hemarshitha

Guide Details

Name	Department	Contact Number
Ms. Siji Rani	Department of Computer Science and Engineering	–

1. Project Title

Project Title:

Predictive Reaction Simulation for Adaptive, Multi-Turn, Multilingual Counterspeech Generation

2. Project Abstract

2.1 Motivation

Motivation:

Hate speech is becoming more common on social media, and platform moderators as well as NGOs resort more and more to counterspeech, which are nonviolent responses to hate speech, instead of removing the content entirely. Recent works have established that it is possible for LLMs to produce counterspeech in a zero-shot way, without any specific training on the task. Yet, the current systems regard this as a one-shot task in that hateful comments are inputted and one response is produced and the process is done. In fact, online discourse is conversational in nature and takes place through multiple turns, and even though the produced response might appear eloquent, it is quite possible that it will make the commentator more hostile instead of defusing the situation. This issue is important as platform moderators, NGOs (e.g., those referenced in the CONAN data set), and regular users alike could all benefit from de-escalatory responses. Additionally, most of the existing literature on counterspeech, including the one that the current project is based on, is solely in the English language, although hate speech is a global phenomenon and can occur in many consequences through code-mixing, especially on Indian social media. The current project has been initiated because there is a need to develop counterspeech generators that can reason about their own impact and can operate in multiple languages.

2.2 Problem Statement

Problem Statement:

Current zero-shot and fine-tuned approaches to counterspeech generation do not have a process in place to predict how likely it is that the generated counterspeech may escalate or de-escalate the ongoing discussion, nor do they adapt their approach to generate counterspeech according to how a discussion may evolve over several turns. In addition, all current systems use English data in order to train and evaluate their models, which restricts the use of these systems in the linguistic diversity and code-mixing observed in real-world conversations. There currently does not exist any system which is able to predict (a) the probable reaction of the hateful user to a counterspeech that was generated, (b) refine its counterspeech based on the prediction made, (c) adapt the process over multiple turns, and (d) be used in multiple languages. This project seeks to fill all of these gaps in the scope of a two-semester B.Tech final year project without the need to collect data and generate conversations beyond existing hate speech datasets and the help of an LLM.

3. Objectives

Objectives:

1. Generate counterspeech through a pipeline which anticipates how the hate-speech will respond to a potential reply prior to generating the actual reply.

2. Create an iterative feedback loop which will refine the counterspeech based on the anticipation of how it might cause the hate-speech to escalate.

3. Extend the iterative refinement process to span several rounds of conversation in which the trend towards escalation is tracked over the simulated dialogue.

4. Evaluating the pipeline in at least two languages (English and an Indian language such as Hindi or Tamil) will determine whether the reaction-simulation paradigm is transferable

beyond English. the generated counterspeech is to be done through automatic measurements (from previous studies: toxicity, quality, argument strength) and through a human subject experiment determining de-escalation chances.

4. Dataset / Data Source (if applicable)

Dataset / Data Source:

Primary datasets (Publicly available): CONAN and CONAN-MT (Chung et al., 2019; Fanton et al., 2021) which include pairs of hate speech/counter speech, with the multilingual subsets (French, Italian) from CONAN-MT as the baseline datasets for conducting evaluations in multiple languages; Reddit and Gab Hate Speech Datasets (Qian et al., 2019) for realistic hate speech (not synthetic). Secondary/Constructed Datasets: Small number of multi-turn conversations created synthetically through LLMs (Explicitly described in our methodology due to lack of multi-turn counterspeech publicly available dataset). Indian Language Hate Speech Datasets (HASOC, MultiFED corpora) to be used for multilingual evaluation purpose.

5. Expected Deliverables

Expected Deliverables:

software that demonstrates the generate → simulate → refine paradigm, adapted to multi-turn dialogue and at least two languages, available via an easy-to-use web interface.

1. Evaluation report that includes both automatic metrics and a user study (approx. 100–150 evaluated examples) comparing one-shot zero-shot generation to the proposed reaction → simulation → generation.

2. A set of multi-turn simulated hate speech and counterspeech dialogues, along with results of human validation of these synthetic dialogues, that can be reused in future work.

6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	On Zero-Shot Counterspeech Generation by LLMs (Saha et al.)	2024	The first comprehensive zero-shot evaluation of GPT-2, DialoGPT, FlanT5 and ChatGPT on counterspeech generation, along with introducing manual, frequency and clustering-based prompting approaches for generating types.
2	CounterGeDi: A Controllable Approach to Generate Polite, Detoxified and Emotional Counterspeech (Saha et al.)	2022	Employs generative discriminators (GeDi) for guiding generation of counterspeech in directions such as politeness, low toxicity, and target emotions without fine-tuning.
3	Self-Refine: Iterative Refinement with Self-Feedback (Madaan et al.)	2023	Illustrates that an LLM model can generate self-critical feedback loops (feedback → refinement cycle) by evaluating and refining its own generated output without any additional training in seven generation tasks.
4	Generative Agents: Interactive Simulacra of Human Behavior (Park et al.)	2023	Illustrates that LLM agents equipped with memory and reflection mechanisms can create realistic simulations of reactions/behavior that one would exhibit in a particular social situation.
5	Generalizable Multilingual Hate Speech Detection on Low-Resource Indian Languages using Fair Selection in Federated Learning (Singh & Thakur)	2024	Proposes a framework for federated learning (MultiFED) for hate speech detection in low-resourced Indian languages.

6.1 Comparison of Related Work

6.2 Novelty / Research Gap Statement

Novelty / Research Gap:

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Saha et al. (2024)	2024	Zero-shot prompting of GPT-2, DialoGPT, FlanT5, ChatGPT; manual / frequency / cluster-centered prompts for type control	CONAN, CONAN-MT, Reddit, Gab	Single-turn only; no prediction of listener reaction; English only; type-control fails for humour/hypocrisy
2	Saha et al. (2022)	2022	Generative discriminator (GeDi)-guided decoding for politeness/detox/emotion control	CONAN, Reddit	Needs discriminator training per attribute; still single-shot generation with no self-correction or reaction awareness
3	Madaan et al. (2023)	2023	Self-Refine: LLM generates feedback on its own output, then rewrites iteratively	General NLP tasks (dialogue, reasoning, code) – not hate-speech specific	Feedback is generic quality critique, not grounded in a simulated social outcome; never applied to counterspeech / de-escalation
4	Park et al. (2023)	2023	Generative agents with memory + reflection to simulate believable human behaviour	Sandbox social simulation (custom environment)	Simulation used for general social behaviour, not for predicting a hostile user's reaction to a specific reply, and not tied to any refinement loop
5	Singh & Thakur (2024)	2024	Federated learning across multilingual/code-mixed Indian-language corpora	Low-resource Indian language hate-speech corpora	Addresses detection only, not counterspeech generation; no generative or dialogue component

Table 1: Comparison of related work across key aspects

There are two aspects common between all five compared works which do not converge in any way: (i) counterspeech generation (Saha et al. (2024), (2022)), which deals with generating a better reply but does not predict the reaction of the hate-speaker to it, and which is evaluated in a one-turn, mostly English environment; (ii) self-refinement and social simulation studies (Madaan et al. (2023), Park et al. (2023)) show that LLMs can provide quality evaluation of their output and social simulations of people's reactions, respectively, but have not yet been applied for predicting the reaction of a hostile person and refining the generated reply based on this prediction. In addition, multilingual hate-speech detection research (Singh & Thakur (2024)) is dedicated to detecting hate speech in low-resource Indian languages but does not cover generation of a response to this hate speech. The novelty of this project is to apply reaction prediction and self-refinement (modifying the mechanism of (2023) by using not the general quality feedback, but the reaction of a simulated listener, similar to (2023)) in combination with multi-turn and multilingual conversational environment — something which is not found in any of the five works mentioned above.

7. Potential Publication Venues

Possible Conferences

NeurIPS, ICML, ICLR, AAAI, IJCAI, KDD, ICDM, SDM, AISTATS, UAI, CVPR, ICCV, ECCV, WACV, BMVC, ACL, EMNLP, NAACL, COLING, ECAI, ICPR, ICIP, ICASSP, COMSNETS, CODS-COMAD, INDICON, ICACCS, ICCCI, IEEE ANTS, COMPUTE, ICACC, TENCON

Possible Journals

IEEE Access, Expert Systems with Applications, Knowledge-Based Systems, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Information Sciences, Pattern Recognition, Neurocomputing, Artificial Intelligence Review, Machine Learning, Neural Computing and Applications, Journal of King Saud University – Computer and Information Sciences, Multimedia Tools and Applications, Computers & Security, Computer Networks, Future Generation Computer Systems, Computer Methods and Programs in Biomedicine, Artificial Intelligence in Medicine, Data & Knowledge Engineering, ACM Transactions on Intelligent Systems and Technology (TIST), ACM Computing Surveys, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Dependable and Secure Computing (TDSC), Scientific Reports, PLOS ONE, Sensors, Electronics, Heliyon

Selected Venues & Justification:

Conferences: ICON (International Conference on Natural Language Processing, India) — India-based NLP venue with a strong track record of accepting hate-speech and counter-speech work. EMNLP (Conference on Empirical Methods in Natural Language Processing) — leading NLP venue that has published prior counterspeech and controllable-generation work. COLING (International Conference on Computational Linguistics) — co-published the base paper this project extends (LREC-COLING 2024).
Journals: IEEE Access — broad scope, fast review, accepts applied NLP/AI systems work. ACM Transactions on Intelligent Systems and Technology (TIST) — strong fit for applied, system-oriented LLM pipelines. Expert Systems with Applications — publishes applied NLP and text-generation systems with practical evaluation.

8. References

[1] P. Saha, A. Agrawal, A. Jana, C. Biemann, and A. Mukherjee, "On Zero-Shot Counterspeech Generation by LLMs," in *Proc. LREC-COLING 2024*, pp. 12443–12454, 2024. https://aclanthology.org/2024.lrec-main.1090/

[2] P. Saha, K. Singh, A. Kumar, B. Mathew, and A. Mukherjee, "CounterGeDi: A Controllable Approach to Generate Polite, Detoxified and Emotional Counterspeech," in *Proc. IJCAI 2022*, pp. 5157–5163, 2022. https://www.ijcai.org/proceedings/2022/716

[3] A. Madaan et al., "Self-Refine: Iterative Refinement with Self-Feedback," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023. https://arxiv.org/abs/2303.17651

[4] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative

Agents: Interactive Simulacra of Human Behavior," in *Proc. 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. https://arxiv.org/abs/2304.03442

[5] A. Singh and R. Thakur, "Generalizable Multilingual Hate Speech Detection on Low Resource Indian Languages using Fair Selection in Federated Learning," in *Proc. NAACL 2024*, pp. 7211–7221, 2024. https://aclanthology.org/2024.naacl-long.400/

[6] Y.-L. Chung, E. Kuzmenko, S. S. Tekiroğlu, and M. Guerini, "CONAN — COunter NArratives through Nichesourcing: a Multilingual Dataset of Responses to Fight Online Hate Speech," in *Proc. ACL 2019*, pp. 2819–2829, 2019. https://aclanthology.org/P19-1271/

[7] M. Fanton, H. Bonaldi, S. S. Tekiroğlu, and M. Guerini, "Human-in-the-Loop for Data Collection: A Multi-Target Counter Narrative Dataset to Fight Online Hate Speech," in *Proc. ACL-IJCNLP 2021*, pp. 3226–3240, 2021. https://aclanthology.org/2021.acl-long.250/

[8] J. Qian, A. Bethke, Y. Liu, E. Belding, and W. Y. Wang, "A Benchmark Dataset for Learning to Intervene in Online Hate Speech," in *Proc. EMNLP-IJCNLP 2019*, pp. 4755–4764, 2019. https://aclanthology.org/D19-1482/

8. Faculty Remarks

As guide, I hereby confirm that I checked the plagiarism and AI content.

Signature with Remarks by Guide	Signature of Project Coordinator
Name of Guide: _____	Name of Project Coordinator: _____